

Chao Shen

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Zhejiang University, Hangzhou, Zhejiang, China 310027

Education

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- Zhejiang University, College of Control Science and Engineering** Sep 2022 – Present
- Ph.D. Candidate in Control Science and Engineering (GPA: 3.90/4.0)
 - Supervisor: [Mingyang Sun](#), Professor at Peking University, Honorary Lecturer at Imperial College London; [Peng Cheng](#), Professor at Zhejiang University
- Peking University, College of Engineering** Jul 2024 – Present
- Visiting Ph.D. Student of Intelligent DEcision-mAking for Low Carbon Energy Systems Laboratory ([IDEAL Lab](#))
 - Supervisor: [Mingyang Sun](#), Professor at Peking University, Honorary Lecturer at Imperial College London
- Huazhong University of Science and Technology, School of Civil and Hydraulic Engineering** Sep 2018 – Jun 2022
- B.Eng. in Hydraulic and Hydropower Engineering (GPA: 3.96/4.0)
 - Supervisor: [Hui Qin](#), Professor at Huazhong University of Science and Technology

Research Interests

My research focuses on the integration of **artificial intelligence with power and energy systems**, with emphasis on **large language models, reinforcement learning, power system dynamic security analysis, optimal topology reconfiguration, and optimal power flow**. I am currently working on foundation-model-based methods for power system modeling, simulation, optimization, and decision-making, as well as domain-specific benchmark construction and evaluation protocols for reliable AI in power systems.

Publications

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- **C. Shen**, K. Zuo, M. Sun, “[Universal Transient Stability Analysis: A Pre-trained Generative Transformer-enabled Power System Dynamics Prediction Framework](#)”, Applied Energy, 2026.
 - **C. Shen**, Z. Guo, X. Wan, Z. Yang, Y. Zhang, W. Huang, J. Song, Z. Zhang, M. Sun, “[ProOPF: Benchmarking and Improving LLMs for Professional-Grade Power Systems Optimization Modeling](#)”, International Conference on Machine Learning (ICML), 2026.
 - **C. Shen**, Z. Guo, K. Zuo, W. Huang, M. Sun, “[LLM-DMD: Large Language Model-based Power System Dynamic Model Discovery](#)”, IEEE Transactions on Industrial Informatics, 2026.
 - **C. Shen**, K. Zuo, M. Sun, “[Physics-Augmented Auxiliary Learning for Power System Transient Stability Assessment](#)”, IEEE Transactions on Industrial Informatics, 2025.
 - **C. Shen**, K. Zuo, M. Sun, “[Physics-Following Neural Network for Online Dynamic Security Assessment](#)”, IEEE Transactions on Power Systems, 2025. *ESI Highly Cited Paper (Engineering)*.
 - **C. Shen**, F. Sun, H. Chen, Y. Lin, C. Guo, M. Sun, “[Carbon-NeuGC: Neural Granger Causality Based Attribution Analysis of Power System Carbon Intensity](#)”, China Automation Congress (CAC), 2024.
 - Z. Zhang, **C. Shen**, X. Wan, J. Song, M. Sun, “[LLM-Guided Safe Reinforcement Learning for Energy System Topology Reconfiguration](#)”, Applied Energy, 2026.
 - K. Zuo, **C. Shen**, P. Cheng, J. Song, M. Sun, “[Probabilistic Robustness Verified Data-Driven Transient Security-Constrained Optimal Power Flow](#)”, IEEE Transactions on Power Systems, 2025.
 - J. Shao, **C. Shen**, Z. Guo, Y. Huang, M. Sun, “[Workload Reshaping and Waste Heat Valorization for Token Cost Reduction in Large Model Datacenters: A Two Stage Forecasting and Optimization Framework](#)”, Applied Energy, Under Review, 2026.
 - Z. Li, Z. Guo, X. Lu, J. Wang, J. Song, **C. Shen**, J. Wu, M. Sun, “[OptArgus: A Multi-Agent System to Detect Hallucinations in LLM-based Optimization Modeling](#)”, NeurIPS, Under Review, 2026.
 - X. Wang, F. Tang, Y. Wei, Y. He, Z. Liu, P. Jiang, H. Wu, Z. Guo, **C. Shen***, “A Novel Method for Differential-Algebraic Dynamic Model Discovery in Power Systems: An LLM-Based Multi-Agent Collaborative Framework”, IET Renewable

Power Generation, Under Review, 2026.

*Corresponding author.

Patents

- M. Sun, **C. Shen**, Z. Guo, Z. Zhang, X. Huang, J. Song, "Method, Apparatus, and Equipment for Constructing Power System Models Based on Large Language Models", CN Patent Application No. 202512030361.1, filed Dec. 30, 2025, Pending.
- **C. Shen**, M. Sun, C. Guo, Y. Lin, F. Sun, Y. Lin, "A Data-Mechanism Joint-Driven Multidimensional Granger Causality Attribution Method for Carbon Emissions", CN Patent Application No. 202311496211.4, Pending.
- **C. Shen**, Y. Ruan, M. Sun, C. Zhao, R. Ma, R. Chen, A Semi-Supervised Driving Range Estimation and Driving Behavior Analysis Method Combined with Mechanism, CN Patent Application No. 202410920530.1, Pending.

Selected Honours and Awards

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| • Web of Science ESI Highly Cited Paper (Engineering) for PFNN | 2026.02 |
| • Outstanding Graduate Student, Zhejiang University | 2025.11 |
| • Second Prize, The China Graduation Contest on Smart-City Technology and Creative Design | 2023.04 |
| • Outstanding Graduates Award of Huazhong University of Science and Technology | 2022.06 |
| • China National Scholarship | 2021.10 |
| • Finalist Award Mathematics Contest In Modeling (<i>Top 2%</i>) | 2021.04 |
| • First Prize, The Chinese Mathematics Competitions (<i>Top 8%</i>) | 2020.01 |

Academic Service

Invited reviewer for IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid, Applied Energy, International Journal of Electrical Power & Energy Systems, and Electric Power Systems Research.